EXPERIMENT 10

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AIM : Implementation of Hyperledger Fabric Network and Asset Management using Chaincode

THEORY:

What is Hyperledger Fabric? - An open-source, enterprise-grade, permissioned distributed ledger framework hosted by the Linux Foundation. Its modular architecture allows customizable consensus, identity management and smart contracts (chaincode) written in general-purpose languages such as Go, JavaScript and Java. Its key characteristics are:

• **Privacy and confidentiality —** participants are known and authenticated, and channels partition data so only authorized members see specific transactions.

• **Scalability and performance** - ordering is separated from validation, giving high throughput for supply chain, finance and healthcare use cases.

• **Modularity -** regulatory needs. plug-and-play components let organizations match their own trust and

Architecture of Hyperledger

• **Consensus layer** - agrees on the order of transactions and confirms that a block's transactions are correct under the endorsement and consensus policies; it works with the smart-contract layer and keeps data consistent across the network.

• **Smart contract layer** - checks that each transaction follows its policy and contract, and rejects invalid ones. In Fabric a smart contract is chaincode (Go, Node.js and others) that runs in a secure Docker container separate from the endorsing peer.

• **Communication layer** - peer-to-peer message transfer between nodes; every non-faulty node eventually receives every submitted transaction. TLS secures communication, with one-way or mutual authentication.

Experiment • **Identity service** - sets up a trusted root and handles enrolment, registration and

revocation of identities. Fabric CA is the certification authority that manages certificates, and access control lists add further permissions.

• **Policy service** manages the endorsement, consensus and group-management policies. Policies are agreed by channel members when the channel is created and can change as it

evolves.

• **Interoperation** - lets separate blockchain instances communicate through common data and transaction standards, which remains a major challenge for the sector.

CODE AND OUTPUT :

Step 1 - Set up WSL and Ubuntu

wsl --version | wsl-1-v | wsl.exe --install ubuntu

WSL 3.0.1.0 has no installed distributions, so Ubuntu is downloaded and installed.

```
C:\Users\heram>wsl --version
WSL version: 3.0.1.0
Kernel version: 6.18.40.1-1
WSLg version: 1.0.79
MSRDC version: 1.2.7214
Direct3D version: 1.611.1-81528511
DXCore version: 10.0.26100.1-240331-1435.ge-release
Windows version: 10.0.26200.9457

C:\Users\heram>wsl -l -v
Windows Subsystem for Linux has no installed distributions.
You can resolve this by installing a distribution with the instructions
below:

Use 'wsl.exe --list --online' to list available distributions
and 'wsl.exe --install <Distro>' to install.

C:\Users\heram>wsl.exe --install ubuntu
Downloading: Ubuntu
[==============            24.3%                           ]
```

Step 2 - Create the default Unix user

```
Create a default Unix user account: Jai
Invalid username. A valid username must start with a lowercase letter or
 underscore, and can contain lowercase letters, digits, underscores, and
 dashes.
Create a default Unix user account: jai
New password:
Retype new password:
passwd: password updated successfully
usermod: no changes
Help improve Ubuntu!
```

Step 3 - Fix Docker permissions

sudo usermod -aG docker $USER | sudo apt install util-linux-extra

docker ps first fails with "permission denied". The user is added to the docker group and util-linux-extra (which provides newgrp) is installed.

```
ved@LAPTOP-ENAG2HIV:~$ docker ps
permission denied while trying to connect to the docker API at unix:///v
ar/run/docker.sock
ved@LAPTOP-ENAG2HIV:~$ sudo usermod -aG docker $USER
[sudo: authenticate] Password:
ved@LAPTOP-ENAG2HIV:~$ newgrp docker
Command 'newgrp' not found, but can be installed with:
sudo apt install util-linux-extra
ved@LAPTOP-ENAG2HIV:~$ sudo apt install util-linux-extra
Installing:
  util-linux-extra

Installing dependencies:
  liblastlog2-2

Summary:
  Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 0
  Download size: 373 kB
  Space needed: 1405 kB / 1025 GB available

Continue? [Y/n] Y
Get:1 http://archive.ubuntu.com/ubuntu resolute/main amd64 liblastlog2-2
 amd64 2.41.3-3ubuntu2 [26.7 kB]
Get:2 http://archive.ubuntu.com/ubuntu resolute/main amd64 util-linux-ex
tra amd64 2.41.3-3ubuntu2 [346 kB]
Fetched 373 kB in 2s (200 kB/s)
Selecting previously unselected package liblastlog2-2:amd64.
(Reading database ... 35908 files and directories currently installed.)
Preparing to unpack .../liblastlog2-2_2.41.3-3ubuntu2_amd64.deb ...
Unpacking liblastlog2-2:amd64 (2.41.3-3ubuntu2) ...
Selecting previously unselected package util-linux-extra.
Preparing to unpack .../util-linux-extra_2.41.3-3ubuntu2_amd64.deb ...
Unpacking util-linux-extra (2.41.3-3ubuntu2) ...
Setting up liblastlog2-2:amd64 (2.41.3-3ubuntu2) ...
Setting up util-linux-extra (2.41.3-3ubuntu2) ...
Processing triggers for man-db (2.13.1-1build1) ...
Processing triggers for libc-bin (2.43-2ubuntu2.3) ...
ved@LAPTOP-ENAG2HIV:~$
```

Step 4 - Install dependencies

sudo apt install jq curl git -y

```
ved@LAPTOP-ENAG2HIV:~$ sudo apt install jq curl git -y
curl is already the newest version (8.18.0-1ubuntu2.4).
curl set to manually installed.
git is already the newest version (1:2.53.0-1ubuntu1).
git set to manually installed.
Installing:
  jq

Installing dependencies:
  libjq1  libonig5

Summary:
  Upgrading: 0, Installing: 3, Removing: 0, Not Upgrading: 0
  Download size: 393 kB
  Space needed: 1179 kB / 1025 GB available
```

Step 5 - Install Node.js

curl -fsSL https://deb.nodesource.com/setup_20.x | sudo -E bash -

```
ved@LAPTOP-ENAG2HIV:~$ curl -fsSL https://deb.nodesource.com/setup_20.x
| sudo -E bash -
sudo: preserving the entire environment is not supported, '-E' is ignore
d
2026-10-08 03:47:20 - Installing pre-requisites
Hit:1 http://archive.ubuntu.com/ubuntu resolute InRelease
Get:2 http://security.ubuntu.com/ubuntu resolute-security InRelease [137
 kB]
Get:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 k
B]
Get:4 http://archive.ubuntu.com/ubuntu resolute-backports InRelease [137
 kB]
Get:5 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Pac
kages [598 kB]
```

Step 6 Clone fabric-samples

git clone
https://github.com/hyperledger/fabric-samples

```
cd fabric-samples
Cloning into 'fabric-samples'...
remote: Enumerating objects: 16058, done.
remote: Counting objects: 100% (2970/2970), done.
remote: Compressing objects: 100% (567/567), done.
remote: Total 16058 (delta 2593), reused 2403 (delta 2403), pack-reused
13088 (from 2)
Receiving objects: 100% (16058/16058), 24.09 MiB | 1.25 MiB/s, done.
Resolving deltas: 100% (9241/9241), done.
ved@LAPTOP-ENAG2HIV:~/fabric-samples$
```

Step 7 - Download the Fabric binaries and Docker images

./install-fabric.sh docker samples binary

The Fabric v2.5.16 binaries are downloaded and then the Docker images for orderer, ccenv, baseos and CA are pulled.

```
ved@LAPTOP-ENAG2HIV:~/fabric-samples$ curl -sSLO https://raw.githubuserc
ontent.com/hyperledger/fabric/main/scripts/install-fabric.sh
chmod +x install-fabric.sh
./install-fabric.sh docker samples binary

Clone hyperledger/fabric-samples repo

==> Already in fabric-samples repo
fabric-samples v2.5.16 does not exist, defaulting to main. fabric-sample
s main branch is intended to work with recent versions of fabric.

Pull Hyperledger Fabric binaries

===> Downloading version 2.5.16 platform specific fabric binaries
===> Downloading:  https://github.com/hyperledger/fabric/releases/downlo
ad/v2.5.16/hyperledger-fabric-linux-amd64-2.5.16.tar.gz
===> Will unpack to: /home/ved/fabric-samples
  % Total    % Received % Xferd  Average Speed   Time    Time     Time   C
urrent
                                 Dload  Upload   Total   Spent    Left   S
peed
  0      0   0      0   0      0      0      0
  0      0   0      0   0      0      0      0
  0      0   0      0   0      0      0      0
    0
  0      0   0      0   0      0      0      0
  1 121.8M   1  2.13M   0      0  1.37M      0   01:28   00:01   01:27
  3 121.8M   3  4.27M   0      0  1.66M      0   01:13   00:02   01:11
  5 121.8M   5  6.10M   0      0  1.69M      0   01:11   00:03   01:08
  5 121.8M   5  7.26M   0      0  1.57M      0   01:17   00:04   01:13
  6 121.8M   6  7.82M   0      0  1.38M      0   01:27   00:05   01:22
  6 121.8M   6  8.13M   0      0  1.21M      0   01:40   00:06   01:34
  7 121.8M   7  9.18M   0      0  1.19M      0   01:42   00:07   01:35 9
  8 121.8M   8 10.06M   0      0  1.15M      0   01:45   00:08   01:37 7
  8 121.8M   8 10.95M   0      0  1.12M      0   01:48   00:09   01:39 7
```

```
Status: Downloaded newer image for ghcr.io/hyperledger/fabric-orderer:2.
5.16
ghcr.io/hyperledger/fabric-orderer:2.5.16
====>  ghcr.io/hyperledger/fabric-ccenv:2.5.16
2.5.16: Pulling from hyperledger/fabric-ccenv
91094e494159: Pull complete
1a3ecb2ae779: Pull complete
385fa033ccf1: Pull complete
d997db9b0914: Pull complete
c32d94060c0f: Pull complete
1f467a066682: Pull complete
d838ef601b3c: Download complete
Digest: sha256:807493f095f65140ad4cd4543a824c9021a848c8c7aace9bec512e854
5e870f1
Status: Downloaded newer image for ghcr.io/hyperledger/fabric-ccenv:2.5.
16
ghcr.io/hyperledger/fabric-ccenv:2.5.16
====>  ghcr.io/hyperledger/fabric-baseos:2.5.16
2.5.16: Pulling from hyperledger/fabric-baseos
9ba18f3569df: Pull complete
a151ebea138c: Pull complete
b52a0845a0f4: Pull complete
cbe99fed15ad: Download complete
Digest: sha256:109b4a15b28235db0dcf6ecaa7ffd41dbeeecee6551b231d8178dcf57
4a9157b
Status: Downloaded newer image for ghcr.io/hyperledger/fabric-baseos:2.5
.16
ghcr.io/hyperledger/fabric-baseos:2.5.16
===> Pulling fabric ca Image
====>  ghcr.io/hyperledger/fabric-ca:1.5.17
1.5.17: Pulling from hyperledger/fabric-ca
18afb8c8d2a4: Downloading  16.78MB/43.49MB
b9649d3f054a: Downloading  18.87MB/49.19MB
0b966e606d9a: Download complete
6f4ebca3e823: Downloading  17.83MB/29.54MB
c36bacd85367: Download complete
```

Step 8
Verify the Docker images

docker images | grep fabric

The baseos, ca, ccenv, orderer and peer images are present.

```
      66.8MB
ved@LAPTOP-ENAG2HIV:~/fabric-samples$ docker images | grep fabric
ghcr.io/hyperledger/fabric-baseos:2.5.16    109b4a15b282        257MB
      83.2MB
ghcr.io/hyperledger/fabric-ca:1.5.17        453a0a727e82        381MB
       123MB
ghcr.io/hyperledger/fabric-ccenv:2.5.16     807493f095f6       1.01GB
       257MB
ghcr.io/hyperledger/fabric-orderer:2.5.16   1d918fc7e1e0        182MB
      50.7MB
ghcr.io/hyperledger/fabric-peer:2.5.16      58979b65744b        232MB
      66.8MB
hyperledger/fabric-baseos:2.5               109b4a15b282        257MB
      83.2MB
```

Step 9 Start the test network and create the channel

./network.sh up createChannel

```
./network.sh up createChannel
Using docker and docker-compose
Creating channel 'mychannel'.
If network is not up, starting nodes with CLI timeout of '5' tries and C
LI delay of '3' seconds and using database 'leveldb with crypto from 'cr
yptogen'
Bringing up network
LOCAL_VERSION=v2.5.16
DOCKER_IMAGE_VERSION=v2.5.16
/home/ved/fabric-samples/test-network/../bin/cryptogen
Generating certificates using cryptogen tool
Creating Org1 Identities
+ cryptogen generate --config=./organizations/cryptogen/crypto-config-or
g1.yaml --output=organizations
org1.example.com
+ res=0
Creating Org2 Identities
+ cryptogen generate --config=./organizations/cryptogen/crypto-config-or
g2.yaml --output=organizations
org2.example.com
+ res=0
Creating Orderer Org Identities
+ cryptogen generate --config=./organizations/cryptogen/crypto-config-or
derer.yaml --output=organizations
+ res=0
Generating CCP files for Org1 and Org2
[+] up 7/7
 ✔Network fabric_test                    Created                    0.1s
 ✔Volume compose_peer0.org2.example.com  Created                    0.0s
 ✔Volume compose_orderer.example.com     Created                    0.0s
 ✔Volume compose_peer0.org1.example.com  Created                    0.0s
 ✔Container peer0.org2.example.com       Started                    0.8s
 ✔Container peer0.org1.example.com       Started                    0.6s
 ✔Container orderer.example.com          Started                    1.1s
CONTAINER ID   IMAGE                                  COMMAND             C
REATED         STATUS                 PORTS
```

Step 10
Deploy the chaincode

```
 -ccn basic -ccp ../asset-transfer-basic/chaincode-javascript -ccl javas
cript
Using docker and docker-compose
deploying chaincode on channel 'mychannel'
executing with the following
- CHANNEL_NAME: mychannel
- CC_NAME: basic
- CC_SRC_PATH: ../asset-transfer-basic/chaincode-javascript
- CC_SRC_LANGUAGE: javascript
- CC_VERSION: 1.0
- CC_SEQUENCE: auto
- CC_END_POLICY: NA
- CC_COLL_CONFIG: NA
- CC_INIT_FCN: NA
- DELAY: 3
- MAX_RETRY: 5
- VERBOSE: false
executing with the following
- CC_NAME: basic
- CC_SRC_PATH: ../asset-transfer-basic/chaincode-javascript
- CC_SRC_LANGUAGE: javascript
- CC_VERSION: 1.0
+ '[' false = true ']'
+ peer lifecycle chaincode package basic.tar.gz --path ../asset-transfer
-basic/chaincode-javascript --lang node --label basic_1.0
+ res=0
Chaincode is packaged
Installing chaincode on peer0.org1...
Using organization 1
+ jq -r 'try (.installed_chaincodes[].package_id)'
+ grep '^basic_1.0:1d4d4455731128138889f87c307bc2b5fbe664c87b24c035bee15c
bcc7f59b629$'
+ peer lifecycle chaincode queryinstalled --output json
+ test 1 -ne 0
+ peer lifecycle chaincode install basic.tar.gz
```

Step 11 - Chaincode committed on the channel

```
+ res=0
2026-10-08 04:14:01.323 UTC 0001 INFO [chaincodeCmd] ClientWait -> txid
[b8dc37b58e486ac71c33739abf2a18af7c241b59e4b58dfd61708df1344858b6] commi
tted with status (VALID) at localhost:7051
2026-10-08 04:14:01.323 UTC 0002 INFO [chaincodeCmd] ClientWait -> txid
[b8dc37b58e486ac71c33739abf2a18af7c241b59e4b58dfd61708df1344858b6] commi
tted with status (VALID) at localhost:9051
Chaincode definition committed on channel 'mychannel'
Using organization 1
Querying chaincode definition on peer0.org1 on channel 'mychannel'...
Attempting to Query committed status on peer0.org1, Retry after 3 second
s.
+ peer lifecycle chaincode querycommitted --channelID mychannel --name b
asic
+ res=0
Committed chaincode definition for chaincode 'basic' on channel 'mychann
el':
Version: 1.0, Sequence: 1, Endorsement Plugin: escc, Validation Plugin:
vscc, Approvals: [Org1MSP: true, Org2MSP: true]
Query chaincode definition successful on peer0.org1 on channel 'mychanne
l'
Using organization 2
Querying chaincode definition on peer0.org2 on channel 'mychannel'...
Attempting to Query committed status on peer0.org2, Retry after 3 second
s.
+ peer lifecycle chaincode querycommitted --channelID mychannel --name b
asic
+ res=0
Committed chaincode definition for chaincode 'basic' on channel 'mychann
el':
Version: 1.0, Sequence: 1, Endorsement Plugin: escc, Validation Plugin:
vscc, Approvals: [Org1MSP: true, Org2MSP: true]
Query chaincode definition successful on peer0.org2 on channel 'mychanne
l'
Chaincode initialization is not required
ved@LAPTOP-ENAG2HIV:~/fabric-samples/test-network$
```

Step 12 - Set the peer environment and initialize the ledger

```
ved@LAPTOP-ENAG2HIV:~/fabric-samples/test-network$ export FABRIC_CFG_PAT
H=$HOME/fabric-samples/config
export PATH=$PATH:$HOME/fabric-samples/bin
export CORE_PEER_TLS_ENABLED=true
export CORE_PEER_LOCALMSPID="Org1MSP"
export CORE_PEER_TLS_ROOTCERT_FILE=$HOME/fabric-samples/test-network/org
anizations/peerOrganizations/org1.example.com/peers/peer0.org1.example.c
om/tls/ca.crt
export CORE_PEER_MSPCONFIGPATH=$HOME/fabric-samples/test-network/organiz
ations/peerOrganizations/org1.example.com/users/Admin@org1.example.com/m
sp
export CORE_PEER_ADDRESS=localhost:7051
ved@LAPTOP-ENAG2HIV:~/fabric-samples/test-network$ peer chaincode invoke
 -o localhost:7050 --ordererTLSHostnameOverride orderer.example.com --tl
s --cafile $HOME/fabric-samples/test-network/organizations/ordererOrgani
zations/example.com/orderers/orderer.example.com/msp/tlscacerts/tlsca.ex
ample.com-cert.pem -C mychannel -n basic --peerAddresses localhost:7051
--tlsRootCertFiles $HOME/fabric-samples/test-network/organizations/peerO
rganizations/org1.example.com/peers/peer0.org1.example.com/tls/ca.crt --
peerAddresses localhost:9051 --tlsRootCertFiles $HOME/fabric-samples/tes
t-network/organizations/peerOrganizations/org2.example.com/peers/peer0.o
rg2.example.com/tls/ca.crt -c '{"function":"InitLedger","Args":[]}'
2026-10-08 04:16:42.921 UTC 0001 INFO [chaincodeCmd] chaincodeInvokeOrQu
ery -> Chaincode invoke successful. result: status:200
ved@LAPTOP-ENAG2HIV:~/fabric-samples/test-network$
```

Step 13 - Query the assets

```
-C mychannel -n basic -c '{"Args":["GetAllAssets"]}'
[{"AppraisedValue":300,"Color":"blue","ID":"asset1","Owner":"Tomoko","Si
ze":5,"docType":"asset"},{"AppraisedValue":400,"Color":"red","ID":"asset
2","Owner":"Brad","Size":5,"docType":"asset"},{"AppraisedValue":500,"Col
or":"green","ID":"asset3","Owner":"Jin Soo","Size":10,"docType":"asset"}
,{"AppraisedValue":600,"Color":"yellow","ID":"asset4","Owner":"Max","Siz
e":10,"docType":"asset"},{"AppraisedValue":700,"Color":"black","ID":"ass
et5","Owner":"Adriana","Size":15,"docType":"asset"},{"AppraisedValue":80
0,"Color":"white","ID":"asset6","Owner":"Michel","Size":15,"docType":"as
set"}]
ved@LAPTOP-ENAG2HIV:~/fabric-samples/test-network$ peer chaincode query
-C mychannel -n basic -c '{"Args":["ReadAsset","asset5"]}'
{"AppraisedValue":700,"Color":"black","ID":"asset5","Owner":"Adriana","S
ize":15,"docType":"asset"}
ved@LAPTOP-ENAG2HIV:~/fabric-samples/test-network$
```

CONCLUSION :

A **Hyperledger Fabric** test network with two organizations and an orderer was successfully implemented on WSL. The **basic asset-transfer chaincode** was deployed on "mychannel", and the ledger was initialized, queried, and extended with a new asset that was then transferred to another owner. The experiment demonstrated **permissioned networks, chaincode deployment, and asset management on a distributed ledger**.